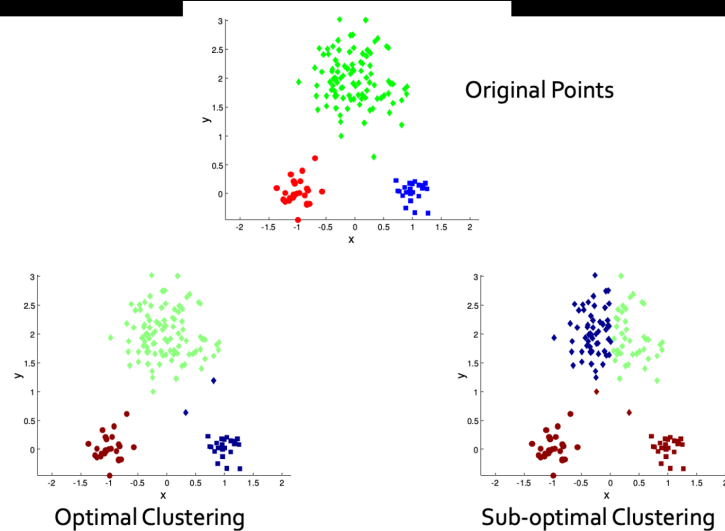


K-Means++

Drawbacks of K-Means

1. K-means is initialization dependent.
 - a. The same data, with different initialization, will get different results (different clusters).

Two different K-means Clusterings



Link: [visualization tool](#) to see this problem

K-Means++

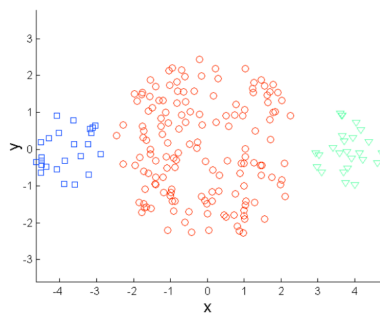
- It uses a smarter way to initialize the centroids to improve the clustering algorithm.
 - Consider data where we want to initialize 3 centroids.
 - We pick the first centroid at random
 - Now, to pick the second centroid, we want to pick a point that is as far away as possible
 - We would want to pick a point that is far away because if two centroids are closer to each other, two clusters for that region of data points will be formed
 - We compute the distance from the centroid C_1 of all the data points present in our dataset D such as $D - \{C_1\}$
 - **Risk:** If we select a datapoint as a second centroid with the farthest distance, then an outlier might be picked as a centroid, and we might have a cluster with the centroid C_2 only.
 - **Solution:** Pick a centroid **probabilistically**, instead of picking it deterministically.
 - I.e. The probability of picking a centroid is proportional to the distance from the first centroid C_1 .
 - The steps involved in the initialization of centroids are:
 - Select the first centroid randomly from the data points.
 - Choose the next center as the farthest point (probabilistically) from the first center.
 - The next center would be a data point farthest from both the first and second centers.
 - Repeat steps 2 and 3 until k centroids have been sampled.
1. If there are **outliers** in our data, then instead of choosing them as centroid, we can choose the farthest point as the centroid with a **probability proportional to the distance**. (Default implementation of Sklearn)

Limitation of KMeans/KMeans++

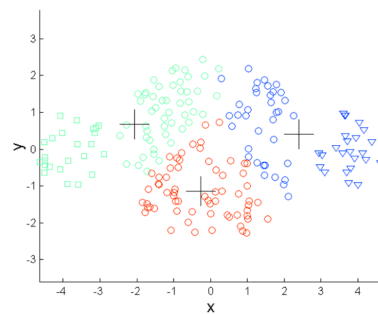
Viz link: <https://hckr.pl/k-means-visualization/>

1. The algorithm may not give the best results for data where the clusters are of varying size or density.

Limitations of K-means: Differing Sizes

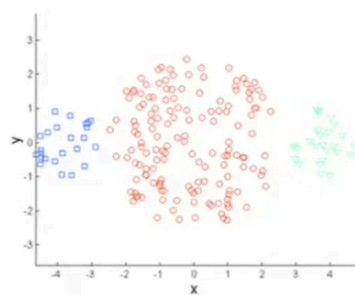


Original Points

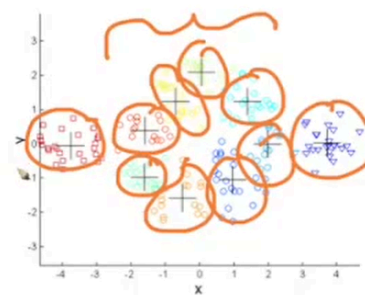


K-means (3 Clusters)

- **How to solve this problem?** increase the value of K.
- Once clusters are formed, similar clusters can be grouped to form a mega cluster.



Original Points



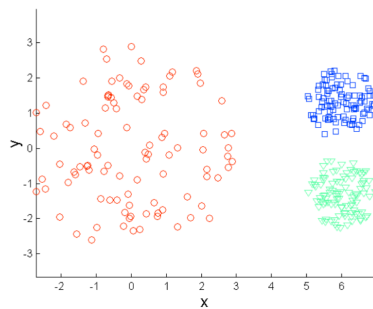
K-means Clusters

One solution is to use many clusters.
Find parts of clusters, but need to put together.

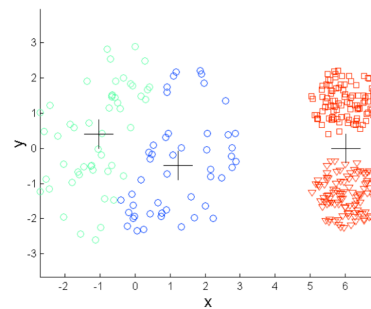
- The problem with this approach is the grouping of similar clusters is not easy

2. It doesn't work well with clusters of different densities.

Limitations of K-means: Differing Density



Original Points



K-means (3 Clusters)

3. It does not work well with non-globular clusters.

Limitations of K-means: Non-globular Shapes

